Introduction to Generative AI

Understanding Foundation Models and LLMs

Core Concepts

What are they and how are they trained?

What are Generative AI Models?

Large Language Models (LLMs) are a specific type of AI within a broader class known as **Foundation Models**.

- These include Language models, Vision models, and Coding models.
- A defining characteristic is versatility: the **same model** can drive any number of tasks and perform multiple different functions.
- They achieve this capability by being trained on a huge amount of unstructured data, often amounting to terabytes of information.

Abstract glowing artificial intelligence brain neural network

How are LLMs Trained?

Exposure

Models are initially exposed to **vast datasets**, ingesting massive amounts of unstructured text from across the internet to build a foundational vocabulary.

Pattern Recognition

Through complex algorithms, the model learns grammar, facts, and reasoning abilities via intricate **pattern recognition** within the data.

Fine-tuning

The base model undergoes rigorous **fine-tuning and iteration** to optimize it for specific tasks, improve safety, and align it with human preferences.

How Do You Work With an LLM?

Input Prompt: You provide instructions, context, or a question to the model.

Tokenization: The model breaks down your input into smaller units called tokens for processing.

Contextual Understanding: The AI analyzes the relationships between tokens to grasp the meaning.

Prediction: It statistically predicts the most likely next token based on its training.

Output Generation: The final synthesized response is delivered back to you.

Impact & Capabilities

Benefits, Limitations, and Applications

Advantages & Benefits of Gen AI

Efficiency & Output

- **Performance Boosts:** Capable of handling complex analytical tasks rapidly.
- **Productivity Gains:** Automates repetitive workflows across various domains.
- **Time Saving:** Drastically reduces the time required for drafting and research.

User Experience & Support

- **Personalization:** Tailors content and responses to individual user needs.
- **Creative Support:** Acts as a brainstorming partner to overcome blocks.
- **Accessibility:** Makes complex information easier to understand.
- **Learning Aid:** Provides explanations and tutoring on diverse subjects.

Disadvantages & Limitations

Technical & Operational Constraints

- **Compute Costs:** Foundation models are notoriously expensive to train and run.
- **Context Gaps:** Models may struggle with highly specific or recent information not in their training data.

Trust & Reliability Issues

- **Accuracy Issues:** Prone to "hallucinations" or confidently stating incorrect facts.
- **Bias in Outputs:** Can reflect and amplify biases present in their training data.
- **Over-reliance:** Risk of users accepting outputs without critical human review.
- **Privacy Concerns:** Issues surrounding the data used for training and user inputs.

Common Applications

Content Creation

Drafting emails, articles, and generating code.

Problem Solving

Data analysis, debugging, and strategic planning.

Idea Generation

Brainstorming concepts and personalized recommendations.

Ethics in AI

Building Trust and Responsible Use

Principles of Responsible Use

Fairness: Ensuring models do not discriminate and that outcomes are equitable for all users.

Explainability: Striving to understand how models arrive at their conclusions and predictions.

Robustness: Building systems that are reliable, secure, and perform consistently under stress.

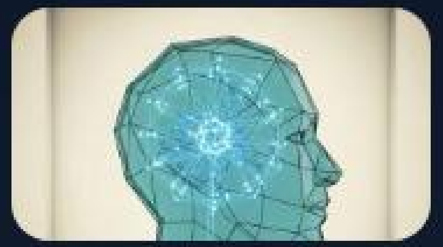
Transparency: Being open about when AI is being used and its known limitations.

Privacy: Safeguarding user data and ensuring information is handled securely and ethically.

Questions?

Thank you for participating.

Image Sources



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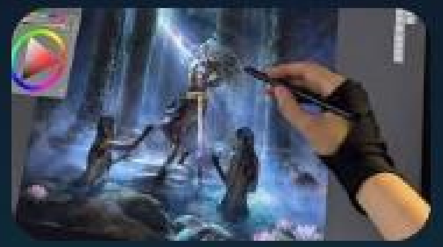

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